

EXTENSION ERP • LG CUSTOMIZATION

Training Manual

CRM, Sales, Contracts & AMC Service Platform

A complete, step-by-step guide to using the LG system — covering customers, deals, quotations, contracts, billing, warranty & AMC service, the management dashboard, and bulk data tools.

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1. Introduction & How to Use This Manual

This manual documents the **LG** system — the platform that runs the complete air-conditioning sales, warranty, and Annual Maintenance Contract (AMC) business: from the first customer enquiry, through quotation approval, contract signing, invoicing and payment collection, to ongoing warranty/AMC service and contract renewal.

It is written for the people who use the system day to day — sales executives, branch/regional managers, service coordinators, call-centre agents, finance/billing staff, and dealers/technicians who interact with it indirectly through email links — rather than for developers. Wherever the underlying system does something automatically (an email, a status change, a calculation), this manual calls it out explicitly so you know what to expect and don't mistake automation for a bug.

HOW SECTIONS ARE ORGANISED

The manual follows the natural order of the business process: Customer → Deal → Quotation → Contract → Invoice → Payment → Renewal, followed by the service/support side (Warranty, AMC, Tickets), then the management dashboard, bulk tools, and a troubleshooting/glossary reference at the end. If you only handle one part of the process (e.g. you are a service coordinator, not a salesperson), jump straight to Section 10.

CALLOUT BOXES USED THROUGHOUT

Blue boxes are informational notes. **Amber boxes** (like this one) flag something you must do correctly or a step that is easy to get wrong. **Red boxes** flag a known system quirk or limitation — something the system does that looks like it should work differently, documented so you aren't caught off guard.

1.1 What's New in Version 1.1

If you trained on Version 1.0 of this manual, these are the changes in how the system behaves. Each one is explained in full in the section listed.

Area	What changed	Section
Customer ID	New customers are numbered CUS-<Region>-<Branch>-00001 , with a separate counter per region and branch. Region and Branch are now required before a customer can be saved.	4
Deal status	A Deal's status now moves forward automatically as its Quotation goes through the workflow, and Probability is set from the status. Status order is now Qualification → Proposal/Quotation → Negotiation → Spec-In → Won/Award.	5
Warranty/AMC expiry follow-up	Starting 2 months before a contract's AMC or a deal's warranty expires, the AM, then the RSM, then HO get a task, a system notification and an email. Unengaged deals are marked Lost 3 months after expiry (this was 6 months before).	5, 13.1
Multi-year quotations	Year-wise breakup table, a dedicated multi-year print format, corrected per-year pricing, and Payment Term is now optional.	6
Quotation email	The quote is emailed to the customer once , when it moves to Customer Approval Pending. It no longer re-sends on every save.	6
Government customers	The extra grace period for government customers has been removed from the advance-payment and billing follow-ups. All customers now follow the same 30/60-day rules.	6, 7
Work Order	New Work Order tab on the Contract and a Send Work Order button that emails the SSD WO PDF to the branch head.	7
Payment Uploader	Picks up instalments due up to the end of the current month, accepts the Invoice No., and recalculates Outstanding as you edit Amount Received.	8
Visit forms	Dealer visits are now logged row-by-row (Dealer Visit Form), and a new Other Visit record covers visits that are not to a customer or dealer.	10.4
Reports	New Gross Revenue , Net Revenue (with pro-rata revenue) and Get Contract Count By ASM reports. The old Pipeline / AMC Pipeline reports have been removed.	11.1

Branch head reminders	A monthly email of pending billing and payments, and a weekly email + notification listing contracts and customers with mandatory information missing.	13.1
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2. Platform Overview

2.1 What You'll See When You Log In

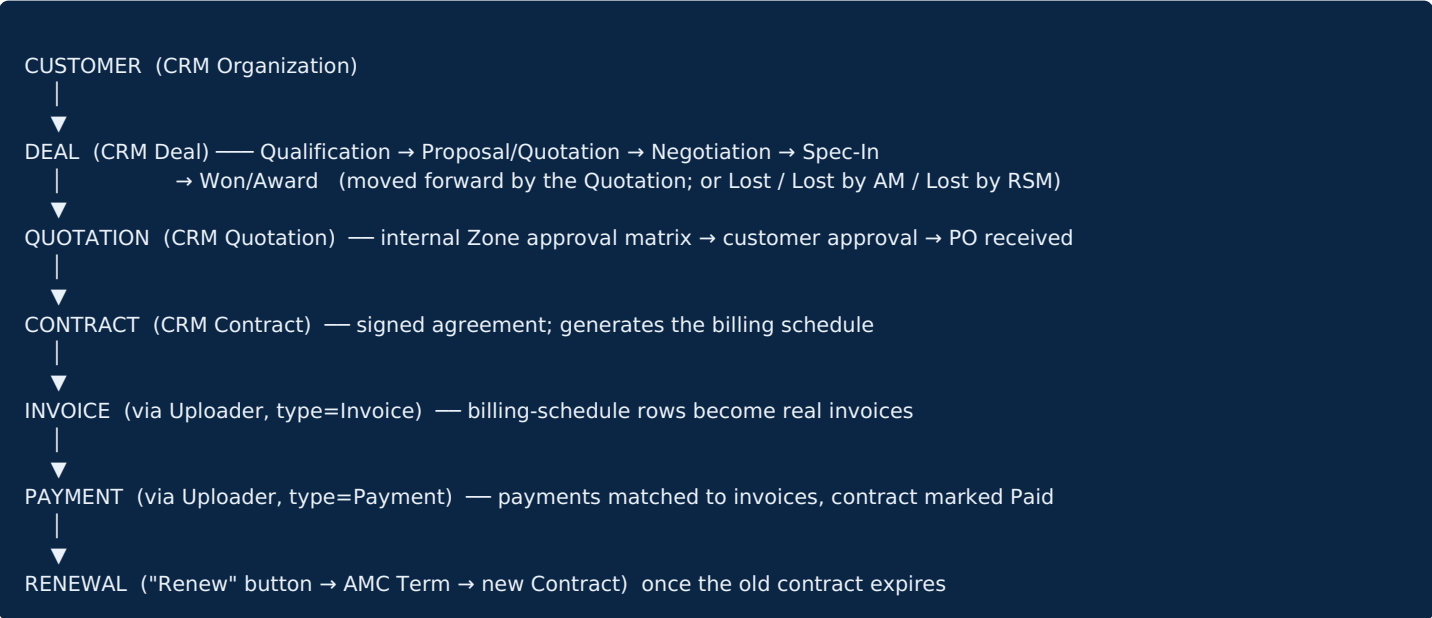
Everything in this manual lives inside one system, reached through your web browser. Day to day, you'll work in two main places:

- **The Desk** — where all your data entry happens: Customers, Deals, Quotations, Contracts, Warranty, AMC, and Tickets are all screens here. Each one gives you a searchable list of records and a form to open, fill in, and save. This is where Sections 3-10 of this manual take place.
- **The Dashboard** — a separate, read-only reporting screen at the `/dashboard` web address, for managers who want revenue/contract charts without opening individual records (Section 11).

You don't need to know how the system is built behind the scenes — this manual describes only what you see and do on screen.

2.2 The End-to-End Business Process

Almost everything in the system is a stage in one pipeline. Learn this flow first — every later section maps onto one box in this diagram:



Running alongside this sales pipeline is the **service side**: Warranty and AMC records, field Technicians and Dealers, and a call-centre Ticket Management screen that logs complaints, repairs, defects, escalations, and spare parts consumed — see Section 10.

2.3 Roles You Will See Referenced

Role / Person	What they typically do in the system
Sales Executive / Account Manager (AM)	Creates and progresses Deals, builds Quotations, is the first approval tier on the zone matrix.
RSM (Regional Sales Manager)	Second-tier approver on higher-value/higher-risk quotations; oversees a region's pipeline via the dashboard.

Zone Approvers (Yellow / Orange / Red)	Special roles that gate quotation approval as risk/discount level increases (Section 6).
Finance	Notified automatically when a dealer accepts a contract; handles billing/Uploader-driven invoicing and payment reconciliation (Section 8).
Service Coordinator / Call-Centre Agent	Logs Warranty and AMC requests, works Ticket Management records, assigns Technicians (Section 10).
Dealer	Channel partner who earns commission on contracts; receives Accept/Reject emails, doesn't need to log in to respond.
Technician	Field service staff; assigned to visits/repairs; tracked as a master record with rates, category and territory.
Branch / Regional / National Manager	Primary users of the Analytics Dashboard (Section 11).
Branch Head	The user set as Branch Head on a Region Branches record. Receives the monthly pending billing & payments email and the weekly missing-information reminder for their branch(es), receives Work Orders sent from contracts in their branch, and can mark a Service deal "Not Interested" (first level, "Lost by AM") (Sections 5, 7, 13.1).
ASM	The User (ASM) set on each Contract. Treated as the AM for that contract's expiry follow-up, and the person the ASM-wise reports group by (Section 11.1).
HO	Head-office users holding the HO role (named Alok in the system). They take over a Warranty/AMC expiry follow-up 1.5 months after the first notice, and give the final "Interested / Not Interested" decision on a deal already marked "Lost by RSM" (Section 5).

What you can see and do is controlled by the **Role(s)** your administrator has assigned you — some people can only view records, others can create/edit them, and some can also Submit or cancel them. A user needs the matching **Yellow / Orange / Red Zone Approver** role to approve or reject a quotation sitting at that specific zone tier, and the **Finance** role to receive the automatic dealer-acceptance notifications described in Section 7. The Dashboard's Region → Branch → Manager drill-down (Section 11) simply follows how branches are assigned to Branch Heads/Area Managers in the Region Branches records — it isn't a separate access lock, so if your organisation needs a stricter "only see your own branch" rule, ask an administrator to set that up for you.

2.4 How to Create Any Record on the Desk

Day-to-day data entry for everything in this manual — Customers, Deals, Quotations, Contracts, Warranty, AMC, Tickets, and all master data — happens through the same simple pattern on the Desk:

1. Use the search bar at the top of the screen and type what you're looking for, e.g. "Deal", "Contract", "Warranty".
2. Click **+ New** from the list that appears, fill in the required fields (marked with a red asterisk), and click **Save**.
3. Some records — Quotation, Contract, Warranty, Purchase Order — need one more step once you're happy with them: click **Submit** to lock the record in and trigger whatever happens next (emails, status changes, a billing schedule being generated, etc.), as described throughout this manual. Once submitted, a record generally can't be edited directly — you'd need to cancel and amend it instead.

Two other screens exist alongside this main workspace: the **Analytics Dashboard** at [/dashboard](#) (Section 11 — for viewing reports only, no data entry) and a technical interface used to connect a mobile app (Section 15, relevant only to administrators). A couple of older, less-used screens also exist at [/frontend](#) and [/lgcrm](#) — unless you've been

specifically told to use one of these, the main Desk and the Dashboard are the two places this manual focuses on.

3. Master Data — Set These Up First

Before running deals through the system, the following reference ("master") records should exist. Getting these right up front avoids re-work later, since Deals, Quotations and Contracts all pull from them.

Territory & Organisation Masters

Master	Purpose
Region Master	Hierarchical sales territory tree (regions can have sub-regions); carries the Region Head used for contract-acceptance notifications.
Region Branches	Branches within a region, each with its own head, address and GSTIN.
Branch	A simpler branch record (id, name, code, head, email) used elsewhere as a lookup.
Company	Legal entity/company tree (for multi-company setups).
Vertical Master	Industry vertical classification (e.g. Hospitality, Education); carries the HP & Zone Details table used to auto-suggest a quotation's pricing zone from total HP.

Product & Pricing Masters

Master	Purpose
Product / CRM Product / Item	The equipment catalogue — AC units and related items, with HSN/SAC codes, UOM, and item group.
Model No	Equipment model catalogue with HP/tonnage and indoor/outdoor classification — linked from equipment line items everywhere (Deal, Contract, Serial No).
Brand / Item Group / CRM Product Group / Product Bundle	Catalogue classification and bundling.
Item Price List / Price List (AMC)	Base pricing by price list and currency.
UOM	Units of measure used across item/product records.
AMC Price Policy	Minimum/maximum rate bands by Zone × HP Range × Project Type, with the approval level each band requires — the reference table behind the quotation zone/approval logic in Section 6.

Payment & Tax Masters

Master	Purpose
Pay Term	The single source of truth for how a bill is split into instalments. Each Pay Term has a table underneath it (Pay Term List) of rows: Invoice Portion (% of the total) and Days (offset from the billing date to the due date). Set these up before creating Quotations — every billing schedule and every invoice split reads from here.
Payment Frequency	Monthly / Quarterly / Semi-Annually / Annually / 100% for Partial Contract — how often billing repeats.
Payment Terms / Payment Terms Template	Generic due-date and discount rules (credit days/months) used on purchasing-side documents.
Mode of Payment, Currency Exchange, Currency Exchange Settings	Payment channel and multi-currency exchange-rate lookups.

Sales/Purchase Taxes and Charges (+ Templates), Tax Category, Item Tax / Item Tax Template	Standard tax configuration for sales and purchase documents.
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People & Channel Masters

Master	Purpose
Dealer	Channel partner master: name, customer type (Residential/Commercial/Individual), commission rate, LCC/ASP/AMC role type, region/branch, and a Dealer Visits log.
Technician	Field engineer master: contact/address, technician type (ASP/Freelancer/Expert), category (A-D), hourly labour rates (under 6 hrs / under 24 hrs / over 24 hrs), assigned dealer, region.
Supplier / Supplier Group	Vendors for purchasing (spares/equipment procurement side).

Other Reference Data

Pincode Master (pincode/city/state/country lookups), Terms and Conditions (standard T&C text blocks for quotations/contracts), Cost Center, and Purchase Order / Purchase Order Item (for procurement of spares/equipment) round out the reference data. These are set-and-forget once configured correctly.

PRACTICAL TIP

Configure Region Master → Region Branches → Dealer → Pay Term in that order before your first live deal. A Deal or Contract created against an incomplete Pay Term will have a wrong or empty billing schedule, which then has to be manually corrected on the Quotation/Contract later.

4. Customers (CRM Organization)

Every Deal, Quotation and Contract belongs to a **Customer**, represented by the **CRM Organization** record.

Key fields

Field	What it's for
Organization Name	Customer/company name.
Customer ID (Naming Series)	The customer's code, generated automatically on first save as CUS-<Region>-<Branch>-##### (e.g. CUS-WEST-MUM-00001). Each region + branch combination has its own running number. You do not fill this in, and it does not change if the customer's branch is edited later.
Region, Branch	Required — the Customer ID is built from them, so a new customer cannot be saved without both. The Branch also decides which Branch Head is reminded about this customer's missing information (Section 13.1).
Customer HC, Vertical	The customer's HC code and industry vertical. Together with Warranty Expiry Date , these are treated as mandatory information: customers missing any of them are listed in the weekly reminder to the branch head.
Site Location City, Project	Site details. Project is free text here (not a link to the Project record).
Expected HP, Expected Ton	Sizing estimates used during early qualification.
Warranty Expiry Date	Customer-level warranty tracking field (separate from the equipment-level dates on individual product lines).
Contracts Information	Read-only history table — automatically gets a new row every time a Contract against this customer is submitted (contract id, start date, expiry date, status). You never edit this by hand.

Step-by-step: creating a customer

1. Search first. Because the system blocks you from saving a duplicate — **the same organisation name + project + branch combination cannot exist twice** — always search before creating a new record, even if the name looks slightly different from what you expect.
2. Fill in **Organization Name**, city, project, **Region** and **Branch**. Also fill in **Customer HC, Vertical** and **Warranty Expiry Date** if you have them; otherwise the customer will appear on your branch head's weekly missing-information list.
3. Save. The **Customer ID** (**CUS-<Region>-<Branch>-#####**) is generated automatically.
4. Add contacts, addresses, and any Customer Visit Form entries as the relationship develops.

EXISTING CUSTOMERS WERE RENUMBERED

Customers created before Version 1.1 carried codes such as **abc-delhi-2026-07-08**. These have been renamed to the new **CUS-<Region>-<Branch>-#####** format, and the contracts and other records pointing at them were updated at the same time. If you have an old customer code written down somewhere, search by customer name instead.

Creating a Deal from a customer

On a saved customer, the **Create AMC Opportunity** button opens a new Deal of type "Warranty AMC Conversion" for that customer. The Deal's **Customer Address** is filled in automatically: the customer's primary address if one is set, otherwise their first billing address, otherwise any address linked to the customer. (As with the Project button in

Section 4.1, this creates a *Deal*, not an Opportunity record.)

DUPLICATE CHECK

If you get an error saving a new customer, it almost always means one already exists with the same name + project + branch. Search for it rather than trying a slightly different name to force it through — that just creates a second, harder-to-find record for the same customer.

Bulk customer import

Use the **Import Excel** action on the Organization list to bulk-load customers (and any attached Customer Visit Form rows) from a spreadsheet. The importer is flexible about column headers — it matches by label, a "scrubbed" version of the label, a known alias, or the literal field name — and it **updates existing customers instead of duplicating them** when a row's customer code (or name+branch) already matches a record. This means you can safely re-upload a corrected version of the same sheet.

The Project Module — Tracking a Customer's Site/Installation

A **Project** is a separate record from the Deal/Contract pipeline — it represents one specific customer installation or site (a customer can have more than one Project if they have multiple sites), and exists to give you a single, always-current answer to "is this site's cover still active?" without having to dig through that site's Deal history.

KEY FIELDS

Field	What it's for
Project ID	The Project's own record name. It must start with the letters "IL" — the system blocks saving and clears the field if you type anything else. New Projects default to "IL" for you to complete.
Customer Reference No., Customer Name	Links back to the CRM Organization; Customer Name is filled in automatically from it.
Vertical, Parent Vertical	Filled in automatically from the customer's industry — useful for filtering Projects by business type.
Status	<i>Read-only.</i> One of IN Warranty / OUT Warranty / AMC Active / AMC Expired — you never set this by hand, it is written automatically (see below).
Expiry Date	The date this site's current cover (warranty or AMC) runs out — also kept in sync automatically.
Current Opportunity	A hidden, read-only link to whichever Deal is currently considered this Project's "live" one.
Region, Branch, Branch Head Name	The servicing territory for this site.
Address / Shipping Address tab	The site's billing and shipping addresses.

HOW STATUS AND EXPIRY DATE STAY IN SYNC WITH DEALS

You don't update a Project's Status or Expiry Date directly — they are pushed onto the Project automatically every time a Deal that's linked to it is saved: if the Deal has a Warranty Expiry Date, that date and the Deal's own Warranty/AMC Status are copied onto the Project; if it has an AMC Expiry Date instead, that's copied instead. A second, more careful check then re-confirms the update only if the Deal you just saved is genuinely the most recently created Deal for that Project.

GOTCHA — THIS IS ALSO THE CAUSE OF THE SECTION 5 PROJECT GOTCHA

The first, simpler sync described above runs on *every* save of a linked Deal, with no check for whether it's the latest one — only the second check looks at recency. In practice this means editing an old, already-closed Deal for that site can still overwrite the Project's displayed Status/Expiry Date, exactly as flagged in Section 5. If a Project's status looks wrong, re-save the customer's actual current Deal for that site to put it right.

TURNING AN EXPIRED PROJECT INTO A NEW DEAL

Once a Project's Expiry Date has passed, a **"Create AMC Opportunity"** button appears on its form. Clicking it opens a new Deal, with the Deal Owner set to you, and the following pre-filled automatically from the Project:

1. Customer and customer address are copied across.
2. Branch and Region are copied from the customer record.
3. The new Deal's type is set based on the Project's current Status: **AMC Expired** → "Lost AMC Conversion"; **OUT Warranty** → "Lost Warranty Conversion".

Review the new Deal, add/confirm its equipment (Product Details), and work it forward through the normal Deal → Quotation → Contract cycle (Section 5 onward) like any other opportunity.

NAMING TRAP — THIS BUTTON DOES NOT CREATE AN "OPPORTUNITY" RECORD

Despite its label, **"Create AMC Opportunity"** creates a **Deal**, not a record in the separate **Opportunity** doctype described in Section 9.1. The two are unrelated features that happen to share similar wording — if you're looking for the customer's actual Deal from this button, check the Deals list, not Opportunities.

5. Deals & Opportunities (CRM Deal)

The **Deal** is where a salesperson tracks an opportunity from first contact through to a signed contract or a lost cause. This is the screen sales staff live in day to day.

Key fields

Field	What it's for
Status	Qualification → Proposal/Quotation → Negotiation → Spec-In → Won/Award (plus Ready to Close, set by hand), or Lost / Lost by AM / Lost by RSM / Sales. Most moves now happen automatically from the linked Quotation (see the table below).
Probability	Set automatically from the Status every time the Deal is saved (Qualification 10%, Proposal/Quotation 40%, Negotiation 60%, Spec-In and Ready to Close 80%, Won/Award 100%, Lost/Sales 0%). You cannot set it by hand.
Deal Category	Service (AMC/warranty-driven; the default) or Sales (new equipment). This choice affects which expiry date matters and which emails/renewal logic apply later — set it correctly at creation.
AMC Expiry Date / Warranty Expiry Date	The dates the whole AMC/warranty status calculation is built on.
Warranty / AMC Status	Calculated automatically — you cannot set this directly. If an AMC expiry date is present: shows "AMC Active" or "AMC Expired". Otherwise, if a warranty expiry date is present: shows "IN Warranty" or "OUT Warranty".
Product Details	Line-by-line list of equipment on this deal (serial no, product, model, HP/tons, warranty/AMC type, dates, maintenance status).
Total HP	Calculated automatically after every save by adding up the HP column across all Product Details rows.
From Contract	Set automatically when this Deal originated from a contract renewal.
Project	Link to the Project record grouping this customer's site history.

Step-by-step: working a deal

1. Create the Deal from (or linked to) the Customer record.
 2. Choose **Deal Category** — Service or Sales — first; this is not easily changed later.
 3. Add every piece of equipment involved under **Product Details**, with serial number, model, and HP/tons filled in for each row.
 4. When you are ready to quote, click **Create Quotation** and build the **CRM Quotation** (Section 6). A Deal still at Qualification moves to Proposal/Quotation.
 5. From here you do not need to move the Status yourself — it follows the Quotation's workflow (table below).
 6. Once a Contract is created and submitted from the quotation, the Deal is **automatically** updated with the final HP, AMC expiry date and annual revenue — you do not need to re-enter these.

How the Quotation moves the Deal's status

Quotation workflow state	Deal status becomes
Open, any approval stage, Approved	Proposal/Quotation
Customer Approval Pending (quote sent to customer)	Negotiation
PO Pending / PO Received (customer accepted)	Spec-In

Quote Won (PO received, quotation submitted)	Won/Award
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FORWARD ONLY

The status only ever moves *forward*. Clicking "Revise Quote", or creating a second quotation for the same Deal, never moves the Deal backwards. A Rejected or cancelled quotation changes nothing, and Deals that are Lost, Lost by AM, Lost by RSM or Sales are never touched by a quotation.

Not Interested / Interested (Service deals)

When a customer does not want to go ahead with a **Service** deal, it is closed through three levels. Each level is asked for a reason:

Who	When the button shows	Result
Branch Head of the deal's branch	Any status other than Lost / Lost by AM / Lost by RSM	Not Interested → status "Lost by AM"
Region Head of the deal's region	Status is "Lost by AM"	Not Interested → "Lost by RSM"; Interested → back to Qualification
HO (Mr Alok)	Status is "Lost by RSM"	Not Interested → "Lost"; Interested → back to Qualification

A deal left at "Lost by AM" or "Lost by RSM" is finalised as Lost automatically 3 months after expiry (see below).

GOTCHA — HP MUST BE A NUMBER

Every row in Product Details must have a numeric value in the HP column. A blank or non-numeric HP on *any* row blocks the whole Deal from saving, because the system tries to sum the column into **Total HP** immediately after save. If a save fails with an HP-related error, check every product row, not just the one you just edited.

GOTCHA — EDITING AN OLD DEAL CAN OVERWRITE PROJECT STATUS

Saving *any* Deal linked to a Project pushes that Deal's status and expiry date onto the Project record — even if it's an old, already-closed Deal and not the customer's current one. If a Project's displayed status looks wrong after someone edits an old Deal for housekeeping, the fix is to re-save the actual current/latest Deal for that project, not to hand-edit the Project.

AUTOMATIC "LOST" CLEAN-UP — NOW 3 MONTHS

Every night, the system marks a Deal as **Lost** once **3 months** have passed since its warranty/AMC expiry (or the expiry of the contract it renews), unless the Deal is engaged — i.e. at Proposal/Quotation, Negotiation, Spec-In, Ready to Close or Won/Award. Pending "Lost by AM" / "Lost by RSM" deals are finalised as Lost at the same point. The Lost reason keeps the AM/RSM reason if one was given; otherwise an automatic reason is recorded. This replaces the old 6-month rule and is the only automatic Lost rule. Before this happens, the AM, RSM and HO are reminded in turn — see Section 13.1.

6. Quotations & the Approval Matrix

The **CRM Quotation** is a formal document — one you Submit once it's approved — with a built-in multi-tier internal approval process before it can be sent to the customer, followed by tracking of the customer's own acceptance and purchase order.

Key fields

Field	What it's for
Zone	Green / Yellow / Orange / Red — a risk/discount tier. Auto-suggested from the deal's industry (via Vertical Master) and total HP against the HP & Zone Details table; you can override it if pricing genuinely differs.
Required Approval	Which approvers must sign off — AM only, AM+RSM, AM+RSM+Mr Alok, or AM+RSM+Mr Alok+MD — driven by the zone.
Start Date, End Date, Payment Term, Payment Frequency, Billing Terms	Contract period and billing cadence. Changing <i>any</i> of these automatically rebuilds the whole billing schedule (see below) — always set these before fine-tuning individual billing rows, not after.
Payment Term	Optional on a quotation (it used to be mandatory) — you can save a quotation before the payment terms are agreed. Fill it in before the billing schedule is finalised, since due dates come from it.
AMC Term (in Years), AMC Term (in Months), Is Multi Year	Read-only, worked out from the Start and End Dates. A quotation longer than one year is automatically flagged Is Multi Year .
Price Rate as per HP	The year-1 rate per HP. Year 1's charge = Total HP × this rate.
Price Rate as Per Year	For multi-year quotations with escalating pricing: a comma-separated list of the total charge for each year from year 2 onward (first value = year 2, second = year 3, and so on). If you give fewer values than there are years, the last value is carried forward to the remaining years. A partial final year is charged pro-rata.
Year Wise Breakup	Filled in automatically for multi-year quotations: one row per year with its period, months, rate per HP and amount. The quotation total and taxes are the sum of these years. Don't edit the rates here — change the price fields above and save.
Contact Email	Where the quotation email goes. If empty, the customer's email on the Organization is used.
Uploaded	Tick this once the customer's PO document is attached — the system will not let the quotation move on from "PO Pending" until this is checked.

The approval workflow

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Open —Submit for Approval→ Approval Pending —Approve→ Approved
Open —Submit for Approval→ Yellow Zone Approval Pending
—Yellow Zone Approve→ Approved
—Yellow Zone Approve→ Orange Zone Approval Pending
—Orange Zone Approve→ Approved
—Orange Zone Approve→ Red Zone Approval Pending
—Red Zone Approve→ Approved
—Reject→ Rejected
—Reject→ Rejected

Approved —Send Quote to Customer→ Customer Approval Pending
Customer Approval Pending —Quote Accept→ PO Pending
Customer Approval Pending —Revise Quote→ Open
PO Pending —PO Receive→ Quote Won (quotation becomes locked/submitted)

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Only users holding the matching **Yellow / Orange / Red Zone Approver** role can approve or reject a quotation sitting at that tier. A higher-risk zone routes through more approval stages before reaching "Approved".

REMARKS ARE MANDATORY AT REJECTION/APPROVAL POINTS

The system blocks you from moving a quotation out of certain states unless you have filled in the matching remarks field first (general **Remarks**, or the zone-specific **Yellow Zone Approve Rejection Remarks / Orange ... / Red ...** field). Always write your justification *before* clicking the workflow action button — the click itself will fail with an error until the remark is present.

Multi-year quotations

For a quotation running more than one year, the billing schedule splits each year's charge across exactly the instalments that fall inside that year — an **Advance** instalment is counted in the period it starts, a **Post** instalment in the period it closes. The quotation prints with the **AMC Offer Quote Multi Year** format (one row per year with period, months, rate per HP, basic and GST), which is chosen automatically in the print view and in the emails. Single-year quotations keep the **AMC Offer quote** format. In both formats the rate column now shows the **Rate/HP**, and the Annexure-C terms printed match the quotation's **AMC Type** (an unrecognised type prints the Comprehensive terms).

The quotation email to the customer

When a quotation moves into **Customer Approval Pending**, the system emails the customer a covering letter with a summary of the quotation and the PDF attached. It goes to the **Contact Email** on the quotation, or the Organization's email if that is empty. The email is sent **once**, on that move — saving the quotation again later does not re-send it. If there is no email address, or the email fails, you are told so, but the workflow step still goes through: fix the address and send it again rather than re-doing the workflow step.

Advance-payment reminders

If a quotation carries an **Advance Amount** and hasn't been submitted yet, the system checks every night for unpaid advance instalments and emails the quotation's owner reminders at **1, 20, 25, 30 and 60 days** overdue. This is the same for all customers — the extra grace period for government customers has been removed. At 60 days overdue the status is set to "Discontinue" automatically — see quirk #3 in Section 16.

Turning a won quotation into a contract

Once a quotation is submitted (typically after "PO Receive" moves it to Quote Won), a **"Create Sub Contract"** button appears on the form. Clicking it generates a new CRM Contract pre-filled with the quotation's customer, dates, amount and billing schedule — you then review and submit the Contract (Section 7).

7. Contracts (AMC & Sales Agreements)

The **CRM Contract** is the signed, binding agreement. Submitting it is what actually starts the billing and service obligations — nothing before this point commits the company to anything.

Key fields

Field	What it's for
Contract Status	Active / Expired / Discontinue. Set automatically to Expired once Expiry Date passes — but the system never overrides a status you've set manually.
Contract Type	Calculated automatically on save: copied from the Deal's type for a brand-new customer relationship, or classified as "AMC Renewable" vs. "Lost AMC Conversion" by comparing today's date against the <i>previous</i> contract's expiry date plus 3 months.
Start Date, Expiry Date	Contract period — the system blocks saving if the start date is after the expiry date.
Dealer (table)	Each dealer sharing commission on this contract, with their commission/contribution rate and their own accept/reject status.
Billing Schedule	The full instalment plan carried over from the quotation (or regenerated the same way).
Product Details	Equipment covered by the contract; has its own bulk Download/Upload CSV template buttons for editing many rows at once.
Customer PO ID, Customer PO Date	Used later to reconcile bulk invoice-Excel uploads against this contract. The PO Date is also the date the ASM report (Section 11.1) counts the contract under.
User (ASM), ASM Name	Required. The ASM responsible for the contract; ASM Name is filled in from the user. This user is the AM for the contract's expiry follow-up (Section 13.1).
Customer HC, AMC Type, Contract Type, Bill To Code, Amount, Billing Schedule	Mandatory information. Any Active contract missing one of these is listed in the branch head's weekly reminder (Section 13.1) until it is filled in.

The Work Order tab

Work-order details have their own **Work Order** tab on the contract:

Field	What it's for
Work Order No., Work order issue date	The work order's number and date. The number is used in the email subject and as the PDF's file name when the work order is sent.
SSD Name (Dealer), SSD Category, SSD Bill to code	The service dealer (SSD) carrying out the work; the SSD Name is printed on the work order.
Additional Terms and Conditions	Free-text terms printed on the work order in addition to the standard ones.
Upload Signature	An image of the authorising signature, printed on the work order.
Work Order Additional Emails	Extra recipients for the work order, separated by commas.

Click **Send Work Order** (available on any saved contract that is not cancelled) to email the work order as a PDF in the **SSD WO** print format. It goes to the **Branch Head** of the contract's branch plus any Work Order Additional Emails. The contract must have a Branch, and that branch must have a Branch Head, or the button stops with a message telling you which is missing.

What happens automatically when you submit a contract

1. The Deal that produced this contract is updated with the final total HP, AMC expiry date (= this contract's expiry date), annual revenue (= this contract's amount), and a "duplicate trigger date" one month before

expiry — used for renewal timing.

2. Every dealer listed in the contract's Dealer table receives an email with a PDF of the contract (print format "AMC Offer") containing **Accept** and **Reject** links they can click without logging in.
3. A new row is appended to the customer's Organization record (**Contracts Information**) recording this contract's id, dates and status.

If a dealer's **Commission Rate** is changed after the contract is already submitted, that specific dealer automatically receives a new Accept/Reject email about the updated commission.

Dealer accept/reject links

These links work without the dealer logging in. Clicking Accept notifies the dealer, the region's Region Head, and everyone holding the **Finance** role. Clicking Reject updates just that dealer's status on the contract. A contract that seems "stuck" is often just waiting on one dealer's response among several, not on the customer.

GOTCHA — AUTOMATIC RENEWAL-DEAL CREATION DOES NOT RUN

Older documentation and some code comments describe the system automatically creating a renewal Deal six months before a contract expires. **This does not currently happen** — do not rely on it. The working renewal path is the manual "Renew" button described in Section 9. Track upcoming expiries proactively (e.g. via the dashboard or a report) rather than waiting for an automatic prompt.

Nightly automated contract housekeeping

Job	What it does
Expire contracts	Any contract past its expiry date that isn't already Expired/Discontinue is automatically set to Expired.
Billing follow-up	Chases overdue, non-advance billing rows; sends an overdue-payment email to the finance team at 30 days, and automatically discontinues the contract at 60 days. The same rule now applies to government and non-government customers.
Advance-payment reminders	Same pattern as the quotation version (Section 6) — reminders at 1, 20, 25, 30 and 60 days, Discontinue at 60 days — applied to the contract itself.
Expiry follow-up	From 2 months before the contract expires, the ASM, then RSM, then HO are given a task, a notification and an email to follow up the renewal (Section 13.1).

8. Billing, Invoicing & Payments

Invoices and payments are **not** created one at a time by editing a form — they are generated in batches using the **Uploader** tool, working directly off a Contract's billing schedule. Think of the Uploader as a "run an import" action rather than a permanent record you go back and edit.

Step-by-step: raising invoices

1. Create a new Uploader record and set its type to **Invoice**.
2. The system fetches the eligible billing-schedule rows and fills the Invoice Uploader table automatically.
3. Review the rows — anything without a valid Contract ID, or that would duplicate an existing invoice, is skipped automatically.
4. **Save the Uploader.** This is the moment the actual invoices are created — nothing is written to the system before you save.

For each row processed, the system creates a new **Invoice** record, splits it into instalment rows using the contract's Pay Term (each portion gets its percentage of the total and a due date offset from the billing date), and marks the originating billing-schedule row as "Invoice Raised", linked to the new invoice.

Step-by-step: recording payments

1. Create a new Uploader record and set its type to **Payment**.
2. The Payment Uploader table fills automatically with every invoice instalment that is **due up to the end of the current month** and not yet fully paid (including partly-paid ones), showing what is still outstanding on each.
3. Enter the **Amount Received** on each row. **Outstanding Amount** is recalculated as you type, so you can see what will be left on that instalment.
4. **Save the Uploader.** For each row, the system finds the matching invoice — by its record ID or by its **Invoice No.** — applies the received amount to the matching instalment that is not yet fully paid, and recalculates the invoice's total paid/outstanding balance and status (Paid, or Partially Paid). A partly-paid instalment comes back on the next Payment Uploader until it is fully paid.
5. The matching Contract Billing Schedule row is updated to reflect the same status, so the contract's payment history and the invoice always agree.

GOTCHA — ONE BAD ROW CAN AFFECT THE WHOLE BATCH

Both Uploader flows process every row in a single all-or-nothing operation. If one row in the middle of a batch fails (wrong contract id, mismatched amount, etc.), the whole save is aborted — but rows processed *before* the failure may already have been written. Always double-check your upload data before saving, and after any failed upload, verify what was actually created rather than assuming nothing happened.

Bulk reconciliation from Excel

On the Contract itself, an administrator can bulk-mark billing rows as paid by uploading an Excel sheet — the system matches each row's **Customer PO ID** against the contract to apply the update, rather than going through the Uploader tool.

9. Renewals

When a contract's status becomes **Expired** or **Discontinue**, a **"Renew"** button appears on the Contract form — but only if no renewal document already references this contract.

Step-by-step: renewing a contract

1. Open the expired/discontinued Contract and click **Renew**.
 2. A new **AMC Term** document opens, pre-filled with the customer and a reference back to the old contract.
 3. Complete the AMC Term's billing details (payment term, frequency, billing terms, amount, start/end dates, billing schedule).
 4. From the AMC Term, generate the new **CRM Contract**. It is automatically pre-filled with the customer, industry, region, branch and company copied from the old contract, plus the payment term, dates, amount, frequency and billing terms entered on the AMC Term.
 5. Review and submit the new contract as normal (Section 7) — this starts a fresh billing/service cycle.

RENEWAL ITSELF IS A MANUAL PROCESS

The system does not create the renewal contract or renewal Deal for you. From 2 months before expiry, the **expiry follow-up** (Section 13.1) gives the ASM, then the RSM, then HO a task, a notification and an email to chase the renewal, but someone still has to act on it. Use those reminders, the dashboard or the **To Be Expired Contracts / Contract Expiry Report** reports to find contracts due for renewal, and use the Renew button.

Tracking renewal & upsell opportunities (Opportunity)

Separate from the Contract's own "Renew" button, **Opportunity** is a dedicated pre-sales record used by sales executives to actively work a pipeline of upcoming warranty-to-AMC conversions and AMC renewals, rather than reacting only once a contract has already expired.

Field	Purpose
Opportunity Type	Warranty or AMC — which kind of renewal/conversion this opportunity tracks.
Stage	Qualified → Requirement Gathering → Proposal Creation → Proposal Sent → Negotiation/Review → Document Sign → Deal Won/Lost.
Opportunity Status	A finer-grained tag: Renewal, Warranty Conversion, Lost Conversion, Chiller, or AMC+TMS.
Expected Value, Owner (SE Name)	Deal size and the sales executive responsible for working it.
Items (table, Opportunity Item)	The specific equipment lines being renewed/converted, including an SVC_TYPE service-code field used for AMC billing classification.

A built-in lookup can pull, for any item code, its Warranty record and flag units that are within **two months of warranty expiry** — this is the intended source list for creating new Opportunities: don't wait for a customer to call in once their warranty has already lapsed, work the list of soon-to-expire units proactively.

10. Service Operations — Warranty, AMC Records & Support

This section covers the post-sale service side of the business: handling warranty and AMC service requests, running the call centre, and tracking technicians, repairs, defects, escalations and spare parts.

10.1 Warranty

A **Warranty** record tracks a warranty claim/registration for a customer's equipment. It is a document you Submit once approved (naming pattern **ID-###**).

Field	Purpose
Warranty Requester	Who raised it — Customer, Dealer, or Technician.
Organization	The customer this warranty belongs to.
Item (table, Warranty Item)	The specific product(s) under warranty, with item code, quantity, and pricing.
Contacts (table, Warranty Contact)	People to notify — supports marking one contact as primary.
Expiry Date, Warranty Status	Tracks when cover ends and its current state (statuses are configurable, each with a colour for quick visual scanning on list views).
Approval Date, Approval Name, Attachment 1/Attachment 2	Sign-off details and supporting document attachments.

Warranty records are typically created, updated, and looked up through the same kind of guided screen used for Deals — searchable by organisation or by a linked contact — and can be formally submitted once approved.

10.2 AMC Records & AMC Term (Renewal Document)

AMC is a standalone service record (distinct from the CRM Contract's AMC billing) — useful for tracking package-based AMC service outside the main contract pipeline, e.g. sold via a dealer/technician channel rather than the sales team.

Field	Purpose
AMC Owner, AMC Activation Date, AMC End Date	Who owns the record and its active period.
AMC Status	Configurable status (colour-coded, like Warranty Status).
AMC Package, Package Base Price, Total Price	The AMC package sold and its pricing.
Technician Name, Technician Type, Technician Owner, Technician Zone	The assigned field engineer for this AMC.
Dealer Name, Dealer Type, Dealer Incentive	The channel partner and their incentive for this AMC sale.
Organization, Contact	Customer and contact list for this AMC.

AMC Term is the document used specifically for the contract-renewal flow (Section 9) — it is not a general AMC record but a short-lived staging document that captures the new billing terms and then converts into a fresh CRM Contract.

10.3 Ticket Management (Call Centre)

Ticket Mangement [sic — this is the actual name in the system] is the call-centre screen: one record captures an entire support interaction end-to-end, from the caller's identity through to parts consumed and the customer's satisfaction rating, rather than being split across several linked documents.

Field group	What's captured
Caller & ticket	Caller Type (Customer/Dealer), Type of Call (Repair/Installation/Amc visit), Status (Open/Closed/Pending/Escalation), ticket stage and owner.
Customer	Name, ID, phone, pincode/state/country, customer type, date of birth.
Dealer assignment	Assigned dealer, dealer type, dealer phone/zone (North/South/East/West), and a re-allocation timestamp/count if the ticket had to be reassigned.
Equipment & warranty	Model number, date of purchase, warranty status and expiry.
Feedback	Overall, service, and communication ratings, plus a feedback timestamp.
Defect/repair	Defect code and category/sub-category, repair end date/time.
Escalation	Escalation source, reason and type; separate approve checkboxes for dealer and technician.
Parts consumed	Part code/name/category, consumed quantity, plus item code/name/category and a Voice of Customer (VOC) free-text field.

Step-by-step: handling a call-centre ticket

1. Agent takes the call and creates a Ticket Management record, capturing caller type and the type of call (Repair/Installation/AMC visit).
2. Look up the customer, confirm equipment details (model, warranty status) — carried over from the customer's product history where available.
3. Assign a dealer/technician zone; if the assignment needs to change later, the re-allocation fields track that history.
4. Record the defect code/category once diagnosed, and any parts consumed during the repair.
5. If the issue needs to go further (unresolved, disputed, or high-severity), set an escalation source/reason/type and route for the relevant approval (dealer, ASM, or technician sign-off).
6. Capture the customer's satisfaction rating and any free-text feedback (VOC) once the ticket is resolved.
7. Move **Status** to Closed once complete.

10.4 Technicians, Dealers & Field Visits

Technicians and Dealers are master records (Section 3) referenced throughout the service flow. **Visit** is a small summary record (first visit, most recent visit, number of chats/visits, average time spent, days visited) of a dealer's or technician's engagement history.

Individual visits are logged in three places, all with the same fields — visit date, **Purpose of Visit** (from the same Deal Type list used elsewhere), **Other Purpose of Visit** (free text when none of the listed purposes fits), remarks, the visitor, and who accompanied them:

Where	Use it for
Customer Visit Form (table on the customer)	Visits to a customer. Visited By is filled in automatically and is read-only. These rows can also be loaded by the Organization bulk-import tool (Section 4).
Dealer Visit Form (the Dealer Visits table on the Dealer master)	Visits to a dealer — one row per visit.
Other Visit (its own record)	Any visit that is not to a specific customer or dealer.

10.5 Repair, Defect, Escalation & Parts

These four areas exist as their own record types with dedicated fields — for example, **Repair** tracks a defect code, category/sub-category, symptom code, repair end date/time, and remarks; **Defect** tracks an item, a defect code

(A/B/C), an attachment, and a Voice of Customer note; **Escalation** tracks source, reason, type, remarks, and separate ASM/dealer/technician approval checkboxes; **Part** tracks part code/name/category, consumed quantity, replacement type and warranty status of the part itself.

HOW THESE ARE USED TODAY

In current day-to-day operation, the fields described above are most commonly filled in **directly on the Ticket Management record** (Section 10.3), which duplicates the same information in one flat screen for call-centre convenience. The standalone Repair/Defect/Escalation/Part records exist as independent masters/logs for anyone who needs to track this data outside a specific ticket (e.g. defect-code reference lists) — check with your service manager which one your team is expected to use for day-to-day logging.

10.6 Serial No — Per-Unit Warranty/AMC Tracking

Every physical installed unit can be tracked individually via a **Serial No** record: serial number, product/model, HP/tonnage, start date, expiry date, days left to expiry, and a **Maintenance Status** of Under Warranty / Out of Warranty / Under AMC / Out of AMC. A **Type** field (N/A / AMC / Warranty) indicates which coverage the record's dates represent. The near-identical **Product Details** table (Section 5) captures the same fields as line items directly inside a Deal or Contract — including a "dummy serial number" option for units not yet individually serialised.

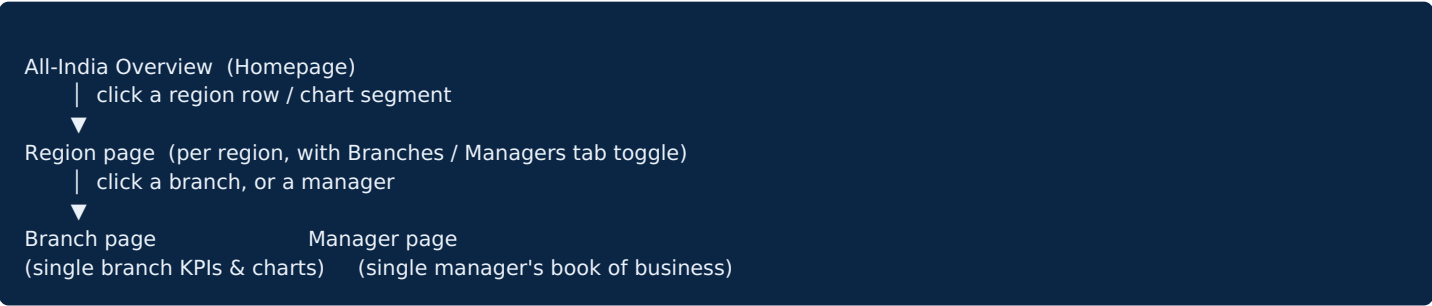
TWO WAYS TO CHECK THE SAME THING

Because Serial No and Product Details share the same fields, a unit's warranty/AMC status can be checked either from the master Serial No list, or by opening the specific Deal/Contract line where that unit was originally sold or contracted. Field/mobile-app users can also look this up per serial number via the Contract API (Section 15).

11. The Analytics Dashboard

The Dashboard is a separate, browser-based reporting app (built with React) aimed at branch managers, regional managers, and executives who need a business overview without opening individual CRM records. It is reached at the site's `/dashboard` address.

Navigation structure (drill-down)



What each level shows

Page	Audience	Contents
All-India Overview	Executives / national management	Four headline KPI cards (Total Revenue with average contract value, Total/Active/Expired Contracts depending on the selected filter, Total Customers, Total Opportunities with quotation count); a stacked bar chart of revenue by region and of contract counts by region, broken down into AMC Renewal / Warranty Conversion / Lost AMC Conversion / Lost Warranty Conversion; a 12-month contract timeline (count and revenue); a pie chart of revenue by industry vertical (click a slice to drill into that vertical's industry breakdown); and a sortable regions table.
Region page	Regional Sales Managers	The same KPI/chart pattern scoped to one region, with an additional toggle between a Branches view and a Managers view for that region, plus donut/area charts for a closer look at composition and trend.
Branch page	Branch Managers	Branch-level KPIs, deal-type donut breakdown, and trend charts scoped to a single branch.
Manager page	Individual Account/Branch Manager review	One manager's personal book of business within their region, using the same chart set scoped to their deals/contracts.

Using the filters

Every page shares a common **Filter Panel** (date range, industry, contract status, and a toggle to use the customer's PO date instead of the contract date for the timeline) plus a global **Reset** to clear back to defaults. Filters you apply persist as you drill down from All-India into a Region, Branch, or Manager, so you don't need to reapply date ranges at each level. Clicking directly on a chart segment (a stacked-bar slice, a pie slice) is a shortcut that both drills down *and* applies the matching filter (e.g. clicking the "AMC Renewal" portion of a region's bar takes you into that region already filtered to AMC Renewal deals).

READING THE FOUR DEAL-TYPE CATEGORIES

Across every chart, contracts/revenue are split into four categories: **AMC Renewal**, **Warranty Conversion**, **Lost AMC Conversion**, and **Lost Warranty Conversion**. These map directly onto the **Contract Type** field described in Section 7 — "Lost" here means the customer lapsed before renewing and this contract represents winning them back, not a failed contract.

11.1 Desk Reports

Alongside the Dashboard, a set of reports is available on the Desk (search for the report's name). Each one has a filter bar at the top and can be exported to Excel from the report's menu. Region, Branch and Category filters on the revenue reports accept several values at once. The three reports below are new in Version 1.1 and are available to the **System Manager**, **Area Manager** and **Region Master** roles.

Gross Revenue — what is owed on raised invoices

Lists every invoice instalment that still has money outstanding as on the **As On Date**, one row per instalment, with the contract, customer, region, branch, ASM, billing details, invoice no., due date, due / received / outstanding amounts, days overdue and an ageing bucket (1–30, 31–60, 61–90, 90+ days). Each row falls into one **Category**:

- **Overdue** — due date has passed and nothing has been received.
- **Partially Paid** — some money has been received but a balance remains.
- **Due This Month** — due later this month and nothing received yet.

Instalments due in a later month with nothing received are not shown. The summary cards show Total Outstanding, Overdue, Partially Paid balance, Due This Month, Overdue 90+ Days, the % collected on these dues, and the number of contracts. Use **Chart By** to chart the outstanding amount by Ageing, Category, Customer, Region, Branch or Due Month.

Net Revenue — billing position and pro-rata revenue by month

Choose a **From Date** and **To Date**. The report shows one group per month, and inside each month the contracts under three cases (filter by **Case** to see just one):

Case	What it shows
Billed	Billing terms that have been invoiced by the end of that month but are not yet fully paid — billed value, received, collected %, and what is left to collect.
Non-Billed	Billing terms whose billing date has arrived (this month or earlier) but whose invoice has not been raised yet.
Revenue	Every contract active in that month with its Active Days and Revenue (Pro-rata) : each billing term's amount spread evenly over the days it covers, so a contract starting on the 12th of a 31-day month earns 20 days' worth, whether or not it has been billed or paid. Contracts without a billing schedule spread their contract value from start to expiry. Revenue stops at the discontinue date, and rejected contracts earn nothing.

Pending amounts carry forward month to month, so the summary cards (Total To Collect, Billed – Awaiting Payment, Due – Not Yet Billed, Collected on Billed, Overdue 90+ Days) show the position at the *end* of the range, while Revenue (Pro-rata) is the total across all months in the range.

Get Contract Count By ASM

Counts **Active** contracts and their revenue per ASM, split by Contract Type (AMC Renewal, Warranty AMC Conversion, Lost Warranty Conversion, Lost AMC Conversion), for each month, quarter or year (**Group By**) of the selected **Year**. Filter by Region, Branch, ASM or Contract Type. An ASM who has contracts in several branches gets one row per branch. Every enabled user with the **Area Manager** role is listed, even with no contracts in the year (as a zero row with no region/branch); those zero rows are left out when a Region or Branch filter is set.

COUNTED BY CUSTOMER PO DATE

This report uses each contract's **Customer PO Date** (not its creation date) to decide which period it belongs to, and it uses the contract's **User** (ASM) field, not the ASM Name text. A contract with no Customer PO Date, or no ASM user, does not appear — fill those in on the contract if a contract seems to be missing.

REPORTS THAT HAVE BEEN REMOVED

The old **Pipeline Report**, **Pipeline Report 1**, **2** and **3**, and the **AMC Pipeline Report** (including its USD versions) have been removed. Use Net Revenue and Gross Revenue instead. Other existing reports — for example Contract Status Report, Contract Expiry Report, To be Expired Contracts, Overdue Billing Report, AR Ageing Report and ASM Wise Contract Comparison Report — are unchanged.

12. Bulk Data Tools (Excel Import/Export)

Several screens support spreadsheet-based bulk operations so staff don't have to hand-enter large volumes of data:

Where	What it does
CRM Organization — Import Excel	Bulk-creates or updates customers and their Customer Visit Form history. Flexible column matching; upserts by customer code (Section 4).
CRM Contract — Download/Upload Product Template	Round-trips the equipment list (Product Details) as a CSV — download to edit many rows in a spreadsheet, then upload to replace the table on the contract.
CRM Contract — bulk invoice reconciliation	Upload an Excel of paid invoices; the system matches rows to contracts by Customer PO ID and marks the corresponding billing-schedule rows Paid.
Uploader (Invoice / Payment)	The primary batch tool for generating invoices and applying payments — see Section 8 for the full step-by-step.

ALWAYS VERIFY AFTER A BULK OPERATION

None of these tools show a live progress bar with per-row confirmation before committing — review the result (the updated list, or the Uploader's processed rows) immediately after each bulk action, and re-run only the rows that didn't go through rather than re-uploading the entire file.

13. Automated Jobs Running in the Background

The following happen automatically, without anyone triggering them. Knowing this list helps you tell "the system did this on its own" apart from "someone changed this manually."

Every night

Job	Effect
Warranty/AMC expiry follow-up	Creates follow-up tasks, notifications and emails as contracts and warranties approach expiry, and marks unengaged deals Lost 3 months after expiry (details in 13.1). This replaces the old "Lost after 6 months" job.
Warranty out / Lost conversion type	A Deal whose warranty has expired (and has no AMC expiry date) is set to "OUT Warranty". On the same 3-month boundary as above, a Deal's or Contract's <i>type</i> becomes "Lost ... Conversion" — this changes the category only, not the status.
Quotation advance-payment reminders	Emails to the quote owner at 1/20/25/30/60 days overdue on an unpaid advance; Discontinue at 60 days. Same for all customers.
Contract advance-payment reminders	Same pattern, applied directly to contracts.
Contract billing follow-up	Overdue-payment email to the finance team at 30 days; auto-discontinues the contract at 60 days. Same for all customers.
Invoice outstanding refresh	Recalculates the outstanding amount on each invoice instalment.
Contract auto-expiry	Any contract past its expiry date is automatically set to Expired (unless already Expired/Discontinue).
Deal status refresh	General daily pass keeping deal status current across the pipeline.

Weekly and monthly

When	Job	Effect
Every Monday, 9:00 AM	Missing-information reminder	Each branch head gets a notification and an email listing the records in their branch(es) with mandatory information missing (details in 13.1).
1st of every month, 9:00 AM	Pending billing & payments	Each branch head gets an email of the billing due this month and the payments still pending in their branch(es) (details in 13.1).

NO AUTOMATIC RENEWAL DEALS

The system no longer creates duplicate/renewal Deals automatically for expired deals. Renewals are chased through the expiry follow-up below and created by hand (Section 9).

13.1 Reminders & Follow-ups You Will Receive

These reminders are addressed to specific people. Emails come from notify.himsolutek@lgepartner.com; a system notification appears under the bell icon on the Desk. They go to whoever is set on the records, so keep the **Branch Head** on each Region Branches record, the **Region Head** on each Region Master, and the **User** (ASM) on each contract up to date. If a branch has no Branch Head, its reminders are skipped and an error is logged for the

administrator.

Warranty/AMC expiry follow-up (daily)

Covers AMC **Contracts** (not Discontinued or Rejected) and warranty **Deals** linked to a Project that have a Warranty Expiry Date but no AMC yet. Responsibility moves up as the expiry date gets closer and then passes:

Stage	When	Who is responsible
AM	From 2 months before expiry, for 1 month	The contract's User (ASM), or the deal's owner — or the branch head if neither is set.
RSM	The next 15 days	The Region Head of the record's region.
HO	From then until 3 months after expiry	Users with the HO role.
Lost	3 months after expiry	Unengaged deals are marked Lost automatically (Section 5).

At the start of each stage, the responsible person gets a **CRM Task** (priority High, due at the end of their stage), a system notification, and an email saying when the cover expires, until when they are responsible, and what happens next. Notices are only sent in the first 7 days of a stage, so nobody is flooded with old stages. The follow-up stops as soon as the renewal deal is Won/Award or Lost.

Pending billing & payments (monthly, 1st at 9:00 AM)

Each branch head receives one email covering all their branches, with two tables:

- **Billing due this month - invoice not raised yet:** billing-schedule rows dated this month that are still Pending, for contracts that are not Discontinued or Rejected.
- **Payments pending:** invoice instalments due this month or earlier that are unpaid or only partly paid, marked Due This Month, Overdue or Partially Paid. These keep appearing every month until fully paid.

Mandatory information missing (weekly, Monday 9:00 AM)

Each branch head receives one system notification with the totals and one email with a clickable list of the records in their branch(es) that are missing mandatory information. Each row shows exactly which fields are missing:

Records checked	Fields that must be filled
Active AMC Contracts (CRM Contract)	Customer HC, AMC Type, Contract Type, Bill To Code, Amount (0 counts as missing), Billing Schedule (at least one row)
All Customers (CRM Organization)	Customer HC, Vertical, Warranty Expiry Date

A record drops off the list as soon as its fields are filled in. The aim is to get the list for your branch to zero.

14. Purchasing & Procurement (Spares/Equipment)

Alongside the sales pipeline, the system has a standard procurement flow for buying spares and equipment from vendors — used by the purchasing/stores team, largely independent of the CRM/Deal side.

Screen	Purpose
Supplier / Supplier Group	Vendor master — name, group, type (Company/Individual/Partnership), default billing currency and default price list. Set up a Supplier before raising any Purchase Order against them.
Purchase Order	The standard "raise a PO to a supplier" document — auto-numbered, and Submitted once approved PUR-ORD-YYYY-#####). Carries a line-item table (item, quantity, UOM, required-by/expected-delivery date, price-list rate), the same Tax Category/Taxes-and-Charges templates used on sales documents, a Payment Terms Template with its own Payment Schedule, and full supplier/shipping/company address details fetched automatically onto the form.
Purchase	A simpler goods-received log line (invoice no./date, item, quantity) with an If Warranty flag — used to record incoming stock that carries a manufacturer warranty, feeding the Warranty/Serial No records described in Section 10.1.

WHO USES THIS

This section is for procurement/stores staff, not sales. It reuses the same Tax Category and Taxes-and-Charges-Template masters described in Section 3 — a template is set up once by finance and then simply selected on each Purchase Order.

15. Mobile & External API (for Administrators)

This section is for whoever administers or troubleshoots the mobile app / external system integration, not for day-to-day CRM users.

The LG app exposes a generic, model-agnostic REST API (under `/api/method/lg.api.master_api.*`) that lets an external client (typically a mobile app) create and read records for any doctype by name, without a dedicated endpoint per doctype:

Endpoint	Purpose
<code>create_record</code>	Create a new record of any model, given a JSON body with a Model name and a Data object of field values.
<code>get_model_detail</code>	Returns the field metadata (labels, types, mandatory flags, options) for a given model — used by the mobile client to build its forms dynamically.
<code>get_record_by_name</code>	Fetch one record by its ID.
<code>get_all_records</code>	Fetch all records of a model (name + display field).

CRM Deal creation is special-cased (an internal `handle_crm_deal` path) with its own stricter, field-by-field validation — mandatory fields, foreign-key/link resolution, and child-table handling are all checked explicitly against a field-definition list, rather than relying purely on Frappe's own doctype-level validation. If a mobile-originated Deal creation fails, check this validation path first rather than assuming it's a generic doctype error.

KEEP THE API CONTRACT IN SYNC

The authoritative request/response contract for this API lives in `apps/lg/lg/api/api_document.md`. Any change to these endpoints should be reflected there at the same time — the mobile app team relies on that document, not on reading the server code.

The dedicated Contract API

Separate from the generic model API above, Contracts have their own purpose-built API used by field/mobile apps: `get_contract` / `get_contracts` (filtered, paginated lookups), `create_contract` / `update_contract` / `delete_contract`, and — notably — `submit_contract` / `cancel_contract`, which let an external app drive a contract's submit/cancel workflow remotely rather than only through the desk UI. The same API also exposes `get_customer`, `get_serial_numbers` (searchable), and `get_serial_details` — so a field technician's app can look up a customer's equipment by serial number and see its warranty/AMC coverage on the spot during a site visit.

Two further admin-only tools worth knowing about: a **bulk Deal export** (System Manager role only) runs as a background job and lets an administrator download the entire Deals list as a file instead of paging through it on screen; and a small **per-user list-column preference** store remembers each user's chosen list-view columns between logins.

16. Known Quirks & Troubleshooting

These behaviours have been confirmed in the live system. None of them are things you did wrong — they are documented here so you recognise them and know the right response.

#	What you'll see	What's actually happening	What to do
1	A quotation moved to "Customer Approval Pending", but the customer says they never got the email.	The email is sent once, on that move, to the quotation's Contact Email (or the Organization's email). If neither is set, or sending fails, you are told so but the workflow step is kept. (Fixed in 1.1: re-saving no longer re-sends the email.)	Fill in the Contact Email and send the quotation again; don't revise and re-approve the quote just to re-send it.
2	An expired contract never automatically produces a renewal Deal, despite this being described in older notes.	The automatic renewal-on-submit logic is dead code and does not run.	Use the manual "Renew" button (Section 9) and track expiries proactively via the dashboard.
3	A Quotation's own overdue-advance-payment breach appears to change the status of a same-named <i>Contract</i> instead of the Quotation.	A known copy-paste issue in the underlying reminder logic. If you notice a Contract's status change without an obvious contract-side cause, check whether a same-named Quotation had an overdue advance payment.	Report the mismatch to your administrator if it causes confusion; don't manually "fix" the Contract status without checking the source Quotation first.
4	A Project's displayed status seems to jump backwards after someone edits an old, unrelated Deal.	Every Deal save pushes its own status/expiry onto its linked Project, regardless of whether it's the customer's current deal.	Re-save the actual current/latest Deal for that project to correct the Project's display.
5	Saving a Deal fails with an error about HP.	Every row in Product Details must have a numeric HP value — the system sums this column immediately after every save.	Check every equipment row for a valid number in HP, not just the row you were editing.
6	A bulk Uploader (Invoice or Payment) fails partway through.	All rows in one Uploader save are processed as a single operation; a bad row can abort the batch after some earlier rows already wrote data.	Verify your source data before saving; after a failure, check what was actually created before re-running.
7	A Contract seems stuck in a pending state even though the customer has clearly agreed.	Contract acceptance also requires each listed Dealer to click Accept on their own email — the contract can be waiting on a dealer, not the customer.	Check the Dealer table's individual statuses before escalating a "stuck" contract.
8	A Contract Billing Schedule row behaves in a way that doesn't match what you can see/edit on the form.	A handful of billing-schedule fields used by the reminder/billing automation (payment date, advance flag, discontinue flag, amount received, portion, outstanding amount, exchange rate, USD amount, "is free") are not exposed as editable fields on the visible form.	Ask an administrator to check the record directly rather than assuming the visible fields are the complete picture.
9	Clicking "Create AMC Opportunity" on a Project opens a Deal, not something in the Opportunities list.	The button's name is misleading — it maps the Project onto a new CRM Deal pre-filled with the customer, branch, region and the correct Lost AMC/Warranty Conversion type. It does not create a record in the separate Opportunity feature (Section 9.1).	Look for the new record in the Deals list, not Opportunities.
10	Saving a new customer fails with "Region and Branch are required to generate the Customer ID".	The Customer ID (CUS-<Region>-<Branch>-#####) is built from the Region and Branch.	Select both, then save.

11	"Send Work Order" stops with "Please select a Branch first" or "Branch Head is not set for this Branch".	The work order is always sent to the branch head of the contract's branch.	Set the contract's Branch, or ask an administrator to set the Branch Head on that Region Branches record.
12	A Deal's status changed to Negotiation, Spec-In or Won/Award without anyone editing the Deal.	Since Version 1.1 the Deal follows its Quotation's workflow automatically (Section 5).	Nothing — this is expected. Check the linked Quotation's workflow state.
13	A contract or customer keeps appearing on the Monday missing-information reminder.	One of its mandatory fields is still empty — for contracts this is often the Billing Schedule (no rows) or an Amount of 0.	Check the "Missing Information" column in the email and fill in exactly those fields.
14	A contract is missing from the Get Contract Count By ASM report.	The report only counts Active contracts that have a Customer PO Date in the selected year and an ASM in the User field.	Fill in Customer PO Date and User on the contract.

17. Glossary

Term	Meaning
AMC	Annual Maintenance Contract — a recurring service agreement covering equipment upkeep.
Deal Category	Service (AMC/warranty-driven) or Sales (new equipment) classification on a Deal.
Zone (Quotation)	Green/Yellow/Orange/Red risk-and-approval tier applied to a quotation based on industry and total HP.
HP	Horsepower — the standard sizing unit for AC equipment used throughout the system for pricing tiers and totals.
Billing Schedule	The instalment plan (dates and amounts) generated from a Pay Term and Payment Frequency, shared by Quotations, Contracts and AMC Terms.
Pay Term	Master record defining how a bill is split into portions (percentages) and their due-date offsets.
Uploader	A one-shot batch tool that turns billing-schedule rows into Invoices, or applies a batch of Payments — see Section 8.
Dealer	Channel partner earning commission on contracts; interacts with the system mainly via email accept/reject links.
Technician	Field service engineer assigned to installations, repairs and AMC visits.
Contract Type — AMC Renewable	A contract renewing an existing relationship in good standing.
Contract Type — Lost AMC Conversion	A contract won back from a customer whose previous AMC had already lapsed more than 3 months earlier.
AMC Term	The short-lived staging document used specifically to capture new terms during a contract renewal (Section 9) — not a general AMC record.
Ticket Management	The single-screen call-centre record covering a support interaction end-to-end (caller, dealer/technician assignment, defect, parts, escalation, feedback).
VOC	Voice of Customer — free-text notes capturing the customer's own words about an issue.
Region Head	The person responsible for a Region Master territory; automatically notified when a dealer accepts a contract in their region.
Opportunity	A pre-sales record for proactively tracking warranty-to-AMC conversions and AMC renewals, separate from the reactive Contract "Renew" button — see Section 9.1.
Serial No	The unique identifier of a specific installed unit; used by field/mobile apps to look up a customer's equipment and its live warranty/AMC coverage on site.
Purchase Order	The procurement document raised against a Supplier for spares/equipment — see Section 14.
Project	A record tracking one customer site/installation's live warranty/AMC status and expiry date, kept in sync from that site's Deals — see Section 4.1. Not to be confused with the "project" text field on CRM Organization.
Customer ID	The customer's code, CUS-<Region>-<Branch>-##### , generated on first save — see Section 4.
Branch Head	The person set on a Region Branches record; receives the branch's work orders and the monthly and weekly reminders — see Section 13.1.
ASM	Area Sales Manager — the User set on a contract; the AM for that contract's expiry follow-up and the person ASM-wise reports group by.
Expiry follow-up	The daily AM → RSM → HO escalation that starts 2 months before a contract or warranty expires — see Section 13.1.
Work Order (SSD WO)	The work order PDF emailed from a contract with the Send Work Order button — see Section 7.

Pro-rata revenue	Revenue counted day by day over the period a billing term covers, whether billed or not — used in the Net Revenue report (Section 11.1).
Create AMC Opportunity (button)	The button on a customer, or on an expired Project, that generates a new renewal/reconversion Deal — despite its name, it does not create an Opportunity record.